화학수학

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연산자

Partial Differential Equations

- General form of a linear second-order PDE

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \cdots, G$ are functions of x and y.

- $G(x, y) = 0$: homogeneous
- $G(x, y) \neq 0$: inhomogeneous

Wave Equation (파동방정식)

From the equation of motion (Newton's law),

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\rho}\frac{\partial^2 y}{\partial x^2}$$

where T is the magnitude of the tension force and ρ is the mass of the string per unit length.

Letting $c = \sqrt{T/\rho}$,

$$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$$

Separation of Variables

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

Assume that $u(x, y) = X(x)Y(y)$.

$X' = \mathrm{d}X/\mathrm{d}x\,, X'' = \mathrm{d}^2X/\mathrm{d}x^2, \cdots$
$Y' = \mathrm{d}Y/\mathrm{d}y\,, Y'' = \mathrm{d}^2Y/\mathrm{d}y^2, \cdots$

If we obtain $f(x, X, X', \cdots) = g(y, Y, Y', \cdots)$, let

$$f(x, X, X', \cdots) = g(y, Y, Y', \cdots) = \lambda$$

where λ is the separation constant.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

Additional Conditions

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

- Boundary conditions

$$y(0, t) = y(L, t) = 0$$

- Initial conditions

$$y(x, 0) = 0, \qquad \left.\frac{\partial y}{\partial t}\right|_{t=0} = f(x)$$

Solving Wave Equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Assume $y(x,t) = \psi(x)\theta(t)$;

$$\psi \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = c^2 \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\theta$$

$$\underbrace{\frac{1}{c^2\theta}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}}_{\text{temporal part}} = \underbrace{\frac{1}{\psi}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}}_{\text{spatial part}} = \lambda$$

Final Solution

- Spatial solution

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

- Temporal solution

$$\theta(t) = b_2 \sin\left(\frac{n\pi c t}{L}\right)$$

- Total solution

$$y(x,t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi c t}{L}\right)$$

The Schrödinger Equation

- 1-dimensional time-dependent Schrödinger equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \mathcal{V}(x,t)\Psi$$

- Consider the case $\mathcal{V}(x,t) = \mathcal{V}(x)$!
- Separation of variables:

$$\underbrace{i\hbar\frac{\theta'(t)}{\theta(t)}}_{\text{depends only on } t} = \underbrace{-\frac{\hbar^2}{2m}\frac{\psi''(x)}{\psi(x)} + \mathcal{V}(x)}_{\text{depends only on } x}$$

Mathematical Operators (연산자)

perform mathematical operations on some function

e.g. "+3" operator, "×2" operator, "$\mathrm{d}/\mathrm{d}x$" operator

- Typically denoted by a letter with a caret ($\hat{}$)
- Operate on the function on its right

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B}+\hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A}\ ?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to commute

- Commutator (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A}\cdots\hat{A}\ (n\ \text{times})$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$

- Not all operators have inverses (*e.g.* $\hat{0}$)

Today's Class

- Operators
  - Operator algebra and differential equation
  - Eigenfunctions and eigenvalues
  - Operators in quantum mechanics

- Symmetry operators
  - Point symmetry operators
  - Symmetry operators on functions

Operator Algebra and Differential Equation

- Derivative operator $\hat{\mathrm{D}}_x = \mathrm{d}/\mathrm{d}x$

- Homogeneous linear DE with constant coefficients

$$a_n \frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1} \frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1 \frac{\mathrm{d}y}{\mathrm{d}x} + a_0 y = 0$$

$$\left(a_n \hat{\mathrm{D}}_x^n + a_{n-1} \hat{\mathrm{D}}_x^{n-1} + \cdots + a_1 \hat{\mathrm{D}}_x + a_0 \hat{E}\right) y = 0$$

This can be solved by operator algebra

Example 13.7

Using operators, solve the equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0.$$

Using the operator notation,

$$\left(\widehat{\mathrm{D}}_x^2 - 3\widehat{\mathrm{D}}_x + 2\hat{E}\right)y = 0$$

Noting that $\left[\widehat{\mathrm{D}}_x, \hat{E}\right] = 0$, factorization leads to

$$\left(\widehat{\mathrm{D}}_x - \hat{E}\right)\left(\widehat{\mathrm{D}}_x - 2\hat{E}\right)y = 0$$

Hence, the two roots are obtained from

$$\left(\widehat{\mathrm{D}}_x - \hat{E}\right)y = 0$$

$$\left(\widehat{\mathrm{D}}_x - 2\hat{E}\right)y = 0$$

Example 13.7

Using operators, solve the equation

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0.$$

$$(\hat{D}_x - \hat{E})y = 0, \qquad (\hat{D}_x - 2\hat{E})y = 0$$

The two equations are identical with

$$\frac{dy}{dx} - y = 0, \qquad \frac{dy}{dx} - 2y = 0$$

which respectively give

$$y_1 = e^x, \qquad y_2 = e^{2x}$$

Hence, the general solution is

$$y = c_1 e^x + c_2 e^{2x}$$

Eigenfunctions and Eigenvalues
(고유함수와 고윳값)

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an eigenfunction of $\hat{A}$, and a is called an eigenvalue. This equation is called an eigenvalue equation (고윳값 방정식).

(Modified) Example 13.2

Find the eigenfunctions and eigenvalues for the operator $\mathrm{d}/\mathrm{d}x$.

We need to find a function that satisfies

$$\frac{\mathrm{d}f}{\mathrm{d}x} = af$$

Solving this differential equation, one can get

$$f = Ce^{ax}$$

where C is an arbitrary constant.

The eigenfunction is $f = Ce^{ax}$, which corresponds to the eigenvalue a.

Operators in Quantum Mechanics

"For every observable quantity,

there is a corresponding mathematical operator"

- Position operators: $\hat{x} = x,\ \hat{y} = y,\ \hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}, \qquad \hat{p}_y = \frac{\hbar}{i}\frac{\partial}{\partial y}, \qquad \hat{p}_z = \frac{\hbar}{i}\frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\widehat{H} = -\frac{\hbar}{2m}\nabla^2 + \mathcal{V}(x, y, z)$$

Example 13.8

Find an eigenfunction of $\hat{p}_x$.

We need to find a function that satisfies

$$\hat{p}_x f = \frac{\hbar}{i}\frac{\partial f}{\partial x} = af$$

The standard form is

$$\frac{\partial f}{\partial x} = \frac{ia}{\hbar} f$$

Solving this differential equation, one can get

$$f = Ce^{iax/\hbar}$$

which is the eigenfunction of $\hat{p}_x$.

Hermitian Property (에르미트 성질)

- If $\hat{A}$ is Hermitian, it obeys the relation

$$\int \chi^* (\hat{A}\psi)\,\mathrm{d}q = \int (\hat{A}\chi)^* \psi\,\mathrm{d}q$$

  where χ and ψ are arbitrary functions and q represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

Example 13.9

Show that the operator $\mathrm{d}/\mathrm{d}x$ is not Hermitian (functions are bound).

$$\int \chi^*\left(\hat{A}\psi\right)\mathrm{d}q = \int \left(\hat{A}\chi\right)^*\psi\,\mathrm{d}q$$

Consider one dimension whose coordinate is x. If the operator is Hermitian, it obeys

$$\int_{-\infty}^{\infty}\chi^*(x)\left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)\mathrm{d}x = \int_{-\infty}^{\infty}\left(\frac{\mathrm{d}\chi}{\mathrm{d}x}\right)^*\psi(x)\,\mathrm{d}x$$

Integration by parts of the left-handed side leads to

$$\int_{-\infty}^{\infty}\chi^*(x)\left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)\mathrm{d}x = \chi^*(x)\psi(x)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\left(\frac{\mathrm{d}\chi}{\mathrm{d}x}\right)^*\psi(x)\,\mathrm{d}x$$

Example 13.9

Show that the operator $\mathrm{d}/\mathrm{d}x$ is not Hermitian (functions are bound).

Need to show:

$$\int_{-\infty}^{\infty} \chi^*(x) \left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right) \mathrm{d}x = \int_{-\infty}^{\infty} \left(\frac{\mathrm{d}\chi}{\mathrm{d}x}\right)^* \psi(x)\, \mathrm{d}x$$

What we have:

bound functions

$$\int_{-\infty}^{\infty} \chi^*(x) \left(\frac{\mathrm{d}\psi}{\mathrm{d}x}\right) \mathrm{d}x = \cancel{\chi^*(x)\psi(x)\Big|_{-\infty}^{\infty}} - \int_{-\infty}^{\infty} \left(\frac{\mathrm{d}\chi}{\mathrm{d}x}\right)^* \psi(x)\, \mathrm{d}x$$

Hence, the operator is not Hermitian.

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Important Example

Show that the eigenvalues of a Hermitian operator are real.

For a Hermitian operator $\hat{A}$, the eigenvalue equation is

$$\hat{A}f = af$$

Multiplying both sides by f^* and integrating,

$$\int f^*(\hat{A}f)\,\mathrm{d}q = \int f^*(af)\,\mathrm{d}q = a\int |f|^2\,\mathrm{d}q$$

Since the operator is Hermitian, the left-handed side is

$$\int f^*(\hat{A}f)\,\mathrm{d}q = \int (\hat{A}f)^* f\,\mathrm{d}q = \int (af)^* f\,\mathrm{d}q = a^*\int |f|^2\,\mathrm{d}q$$

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Important Example

Show that the eigenvalues of a Hermitian operator are real.

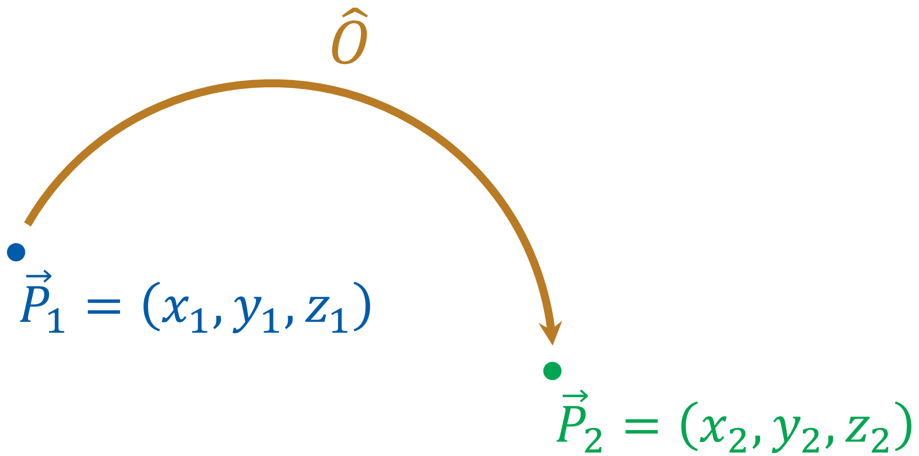
$$a \int |f|^2 \, \mathrm{d}q = a^* \int |f|^2 \, \mathrm{d}q$$

Hence, if $\int |f|^2 \, \mathrm{d}q \neq 0$,

$$a = a^*$$

which means that the eigenvalue a is real.

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators
(점대칭 연산자)

a class of symmetry operators
that leave at least one point unmoved
(the unmoved point = origin)

Identity operator

Inversion operator

Reflection operator

Rotation operator

Identity Operator and Inversion Operator

(항등 연산자 및 반전 연산자)

- Identity operator $\hat{E}$: doing nothing

$$\hat{E}(x_1, y_1, z_1) = (x_1, y_1, z_1)$$

$$\hat{E}\vec{r}_1 = \vec{r}_1$$

- Inversion operator $\hat{\imath}$: inversion w.r.t. the origin

$$\hat{\imath}(x_1, y_1, z_1) = (-x_1, -y_1, -z_1)$$

$$\hat{\imath}\vec{r}_1 = -\vec{r}_1$$

Reflection Operators (반사 연산자) $\hat{\boldsymbol{\sigma}}$

reflection through a plane

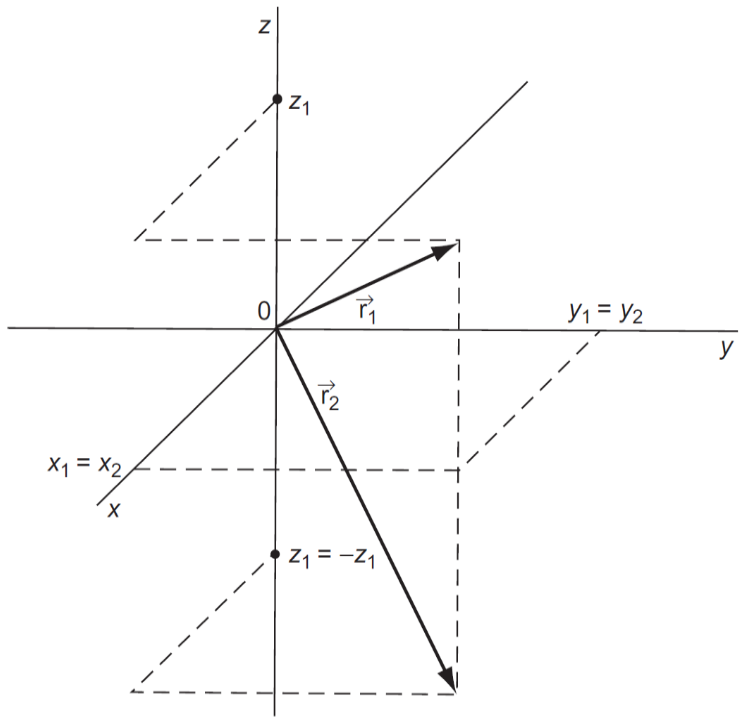
(The plane must be specified)

$$\hat{\sigma}_{(xy)}(x_1, y_1, z_1) = (x_1, y_1, -z_1)$$

$$\hat{\sigma}_{(xz)}(x_1, y_1, z_1) = (x_1, -y_1, z_1)$$

$$\hat{\sigma}_{(yz)}(x_1, y_1, z_1) = (-x_1, y_1, z_1)$$

Action of $\hat{\boldsymbol{\sigma}}_{(xy)}$



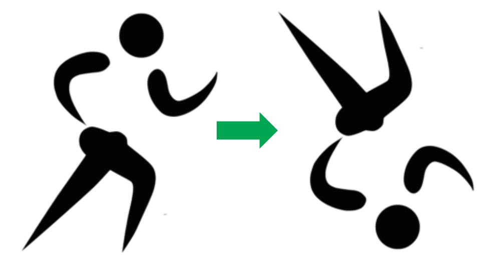
(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.1)

Horizontal and Vertical Reflections

If the system has a principal axis,

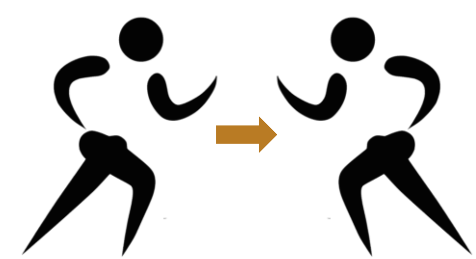
we normally regard it as the z axis

- Horizontal reflection $\hat{\sigma}_{\mathrm{h}} = \hat{\sigma}_{(xy)}$ (unique)



- Vertical reflection $\hat{\sigma}_{\mathrm{v}}$: other reflections

$$\hat{\sigma}_{\mathrm{v}(xz)}, \hat{\sigma}_{\mathrm{v}(yz)}, \cdots$$



Proper Rotation Operators $\hat{\boldsymbol{C}}_{\boldsymbol{n}}$

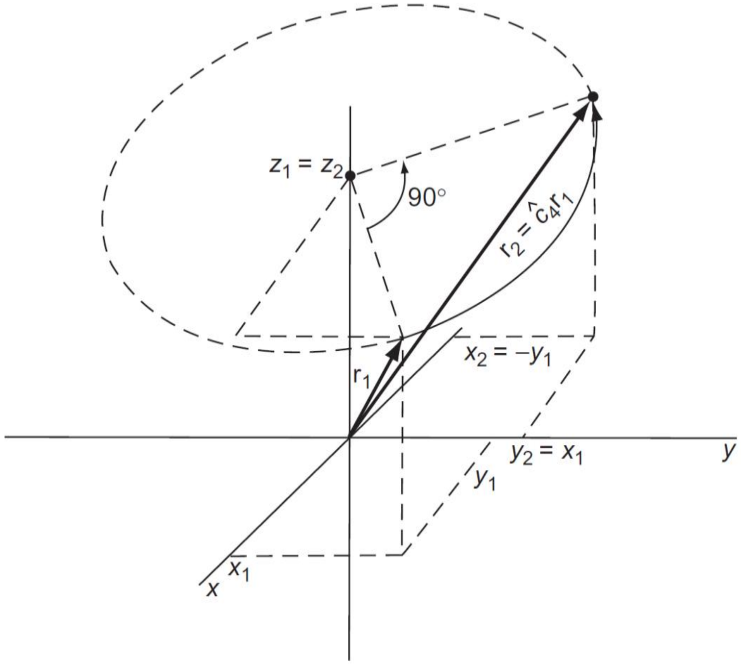
(단순 회전 연산자)

Counterclockwise rotation by $2\pi/n$

- n applications of the rotation operator = 1 full rotation
- The rotation axis must be specified

$$\hat{C}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, z_1)$$

Action of $\hat{C}_{4(z)}$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.2)

Improper Rotation Operators

(반사 회전 연산자)

$\hat{C}_n$ followed by $\hat{\sigma}$

(through a plane perpendicular to the rotation axis)

- The rotation axis must be specified

$$\hat{S}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, -z_1)$$

Examples 13.10 and 13.11

Find $\hat{C}_{4(z)}(1,-4,6)$ and $\hat{S}_{4(z)}(1,2,3)$.

Since $\hat{C}_{4(z)}(x_1,y_1,z_1)=(-y_1,x_1,z_1)$,

$$\hat{C}_{4(z)}(1,-4,6)=(4,1,6)$$

Since $\hat{S}_{4(z)}(x_1,y_1,z_1)=(-y_1,x_1,-z_1)$,

$$\hat{S}_{4(z)}(1,2,3)=(-2,1,-3)$$

Symmetry Elements (대칭 요소)

Point, line, or plane relative to which
the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol

- Inversion i: the origin
- Reflection σ: the reflection plane
- Rotation C_n, S_n: the rotation axis

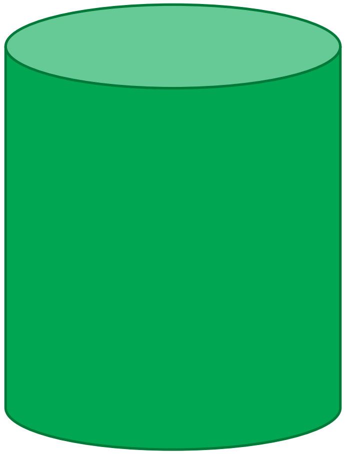
Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, the symmetry operator belongs to the object.

- Coordinate system
  - origin: center of mass of the object
  - z axis: the rotation axis of highest order (largest n)

Example 13.12

List the symmetry operators belonging to a right circular cylinder.



$\hat{\imath}, \hat{C}_{\infty(z)}, \hat{S}_{\infty(z)}, \hat{\sigma}_{\mathrm{h}}, \infty\hat{\sigma}_{\mathrm{v}}, \infty\hat{C}_2, \infty\hat{S}_2$

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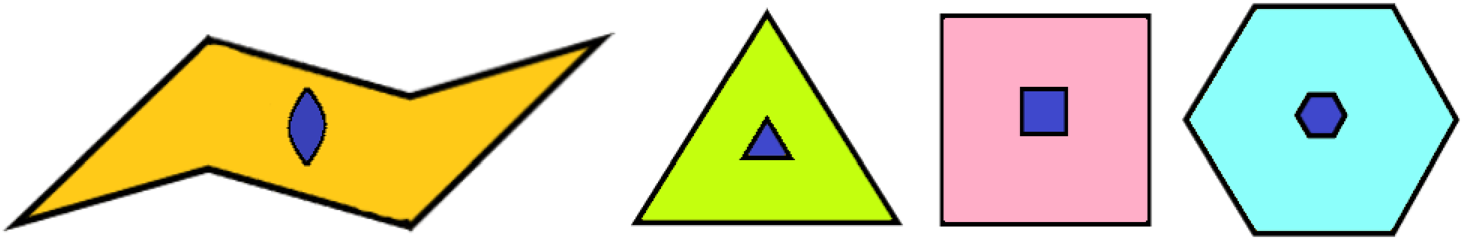
Symbols for Rotational Symmetry

C_2 axis C_3 axis C_4 axis C_6 axis

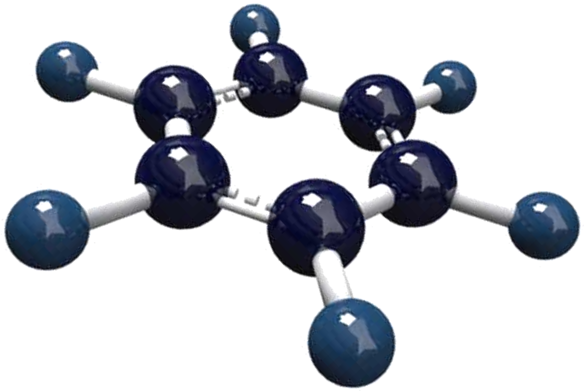


2-fold **3-fold** **4-fold** **6-fold**

(Image: https://bionerdnotes.wordpress.com/2020/06/01/symmetry-operations-and-its-nomenclature-for-the-space-groups-in-crystallography/)

Example 13.13

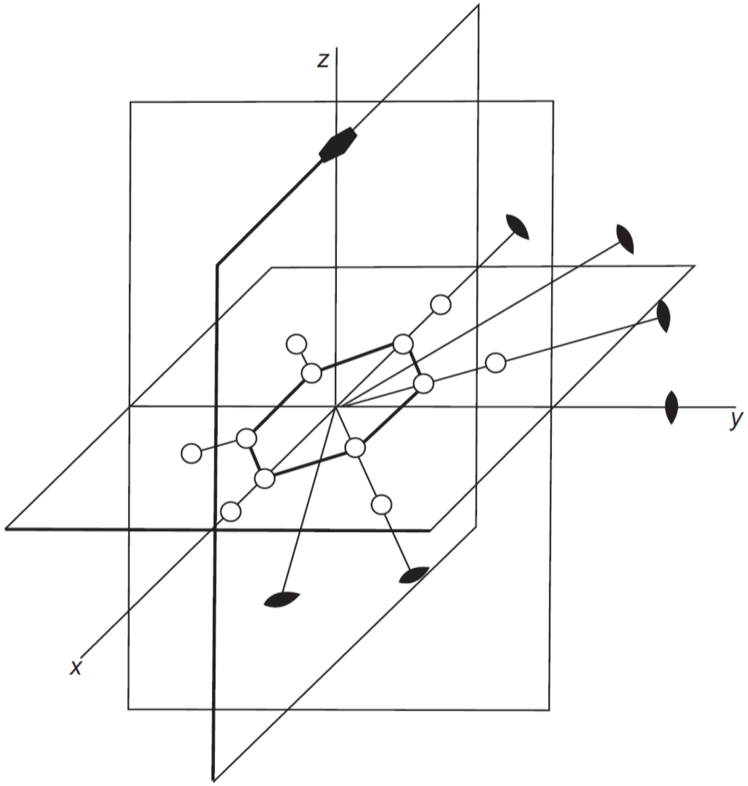
List the symmetry elements of the benzene molecule in its equilibrium conformation.



$\hat{\imath}, \hat{C}_{6(z)}, \hat{S}_{6(z)}, \hat{\sigma}_{\mathrm{h}}, 6\hat{\sigma}_{\mathrm{v}}, 6\hat{C}_2, 6\hat{S}_2$

Example 13.13

List the symmetry elements of the benzene molecule in its equilibrium conformation.



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.3)